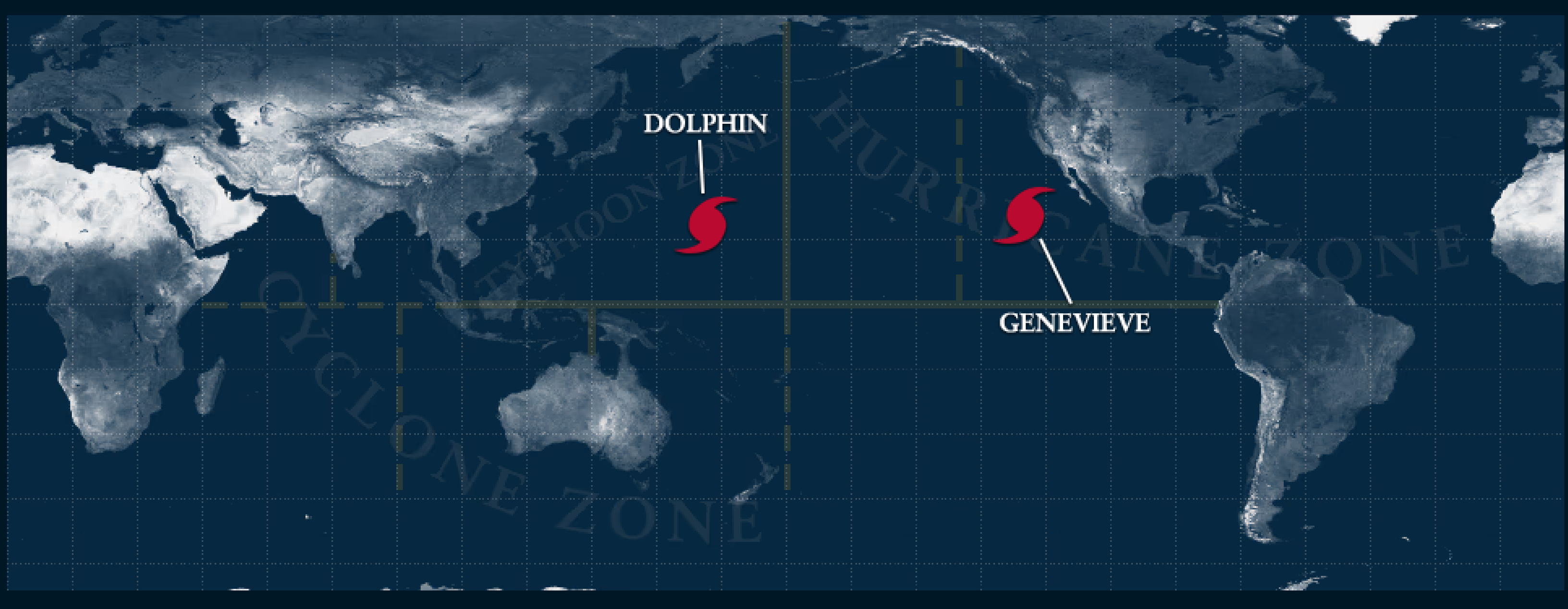
# Stormdom

## Tropical Cyclone Tracking Around the Globe






## Hurricane GENEVIEVE

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Tropical Storm Genevieve Advisory Number  28
NWS National Hurricane Center Miami FL       EP072026
500 PM HST Thu Jul 30 2026

...GENEVIEVE WEAKENS TO A TROPICAL STORM...


SUMMARY OF 500 PM HST...0300 UTC...INFORMATION
----------------------------------------------
LOCATION...21.3N 126.0W
ABOUT 1035 MI...1665 KM W OF THE SOUTHERN TIP OF BAJA C
MAXIMUM SUSTAINED WINDS...70 MPH...110 KM/H
PRESENT MOVEMENT...WNW OR 290 DEGREES AT 15 MPH...24 KM
MINIMUM CENTRAL PRESSURE...989 MB...29.21 INCHES
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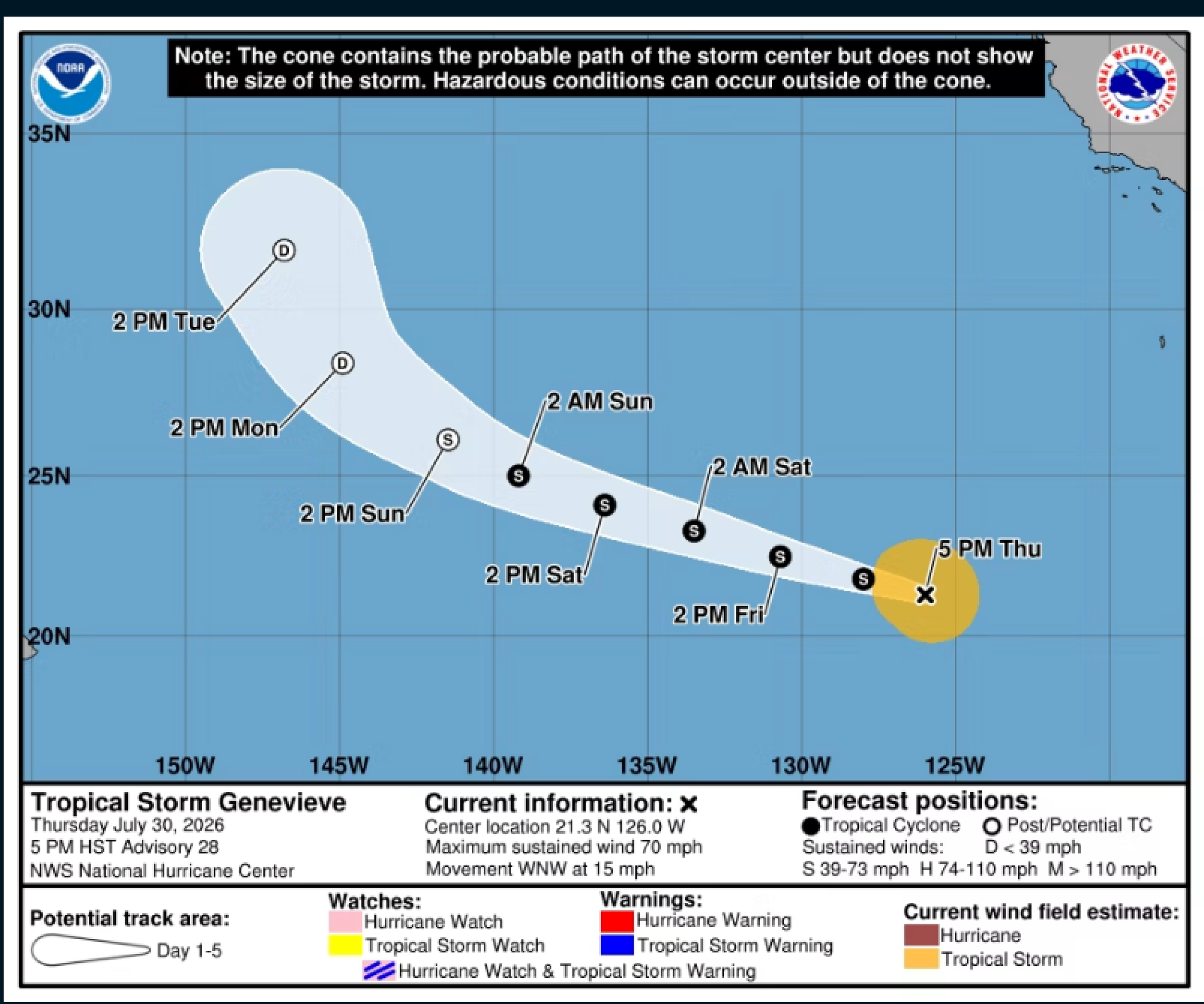


## Super Typhoon DOLPHIN

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1. SUPER TYPHOON 12W (DOLPHIN) WARNING NR 018
   01 ACTIVE TROPICAL CYCLONE IN NORTHWESTPAC
   MAX SUSTAINED WINDS BASED ON ONE-MINUTE AVERAGE
   WIND RADII VALID OVER OPEN WATER ONLY
    ---
   WARNING POSITION:
   310600Z --- NEAR 18.6N 161.0E
     MOVEMENT PAST SIX HOURS - 290 DEGREES AT 09 KTS
     POSITION ACCURATE TO WITHIN 025 NM
     POSITION BASED ON EYE FIXED BY SATELLITE
   PRESENT WIND DISTRIBUTION:
   MAX SUSTAINED WINDS - 140 KT, GUSTS 170 KT
   WIND RADII VALID OVER OPEN WATER ONLY
   RADIUS OF 064 KT WINDS - 050 NM NORTHEAST QUADRANT
                            045 NM SOUTHEAST QUADRANT
                            045 NM SOUTHWEST QUADRANT
                            050 NM NORTHWEST QUADRANT
   RADIUS OF 050 KT WINDS - 080 NM NORTHEAST QUADRANT
                            070 NM SOUTHEAST QUADRANT
                            065 NM SOUTHWEST QUADRANT
                            080 NM NORTHWEST QUADRANT
   RADIUS OF 034 KT WINDS - 150 NM NORTHEAST QUADRANT
                            140 NM SOUTHEAST QUADRANT
                            120 NM SOUTHWEST QUADRANT
                            145 NM NORTHWEST QUADRANT
   REPEAT POSIT: 18.6N 161.0E
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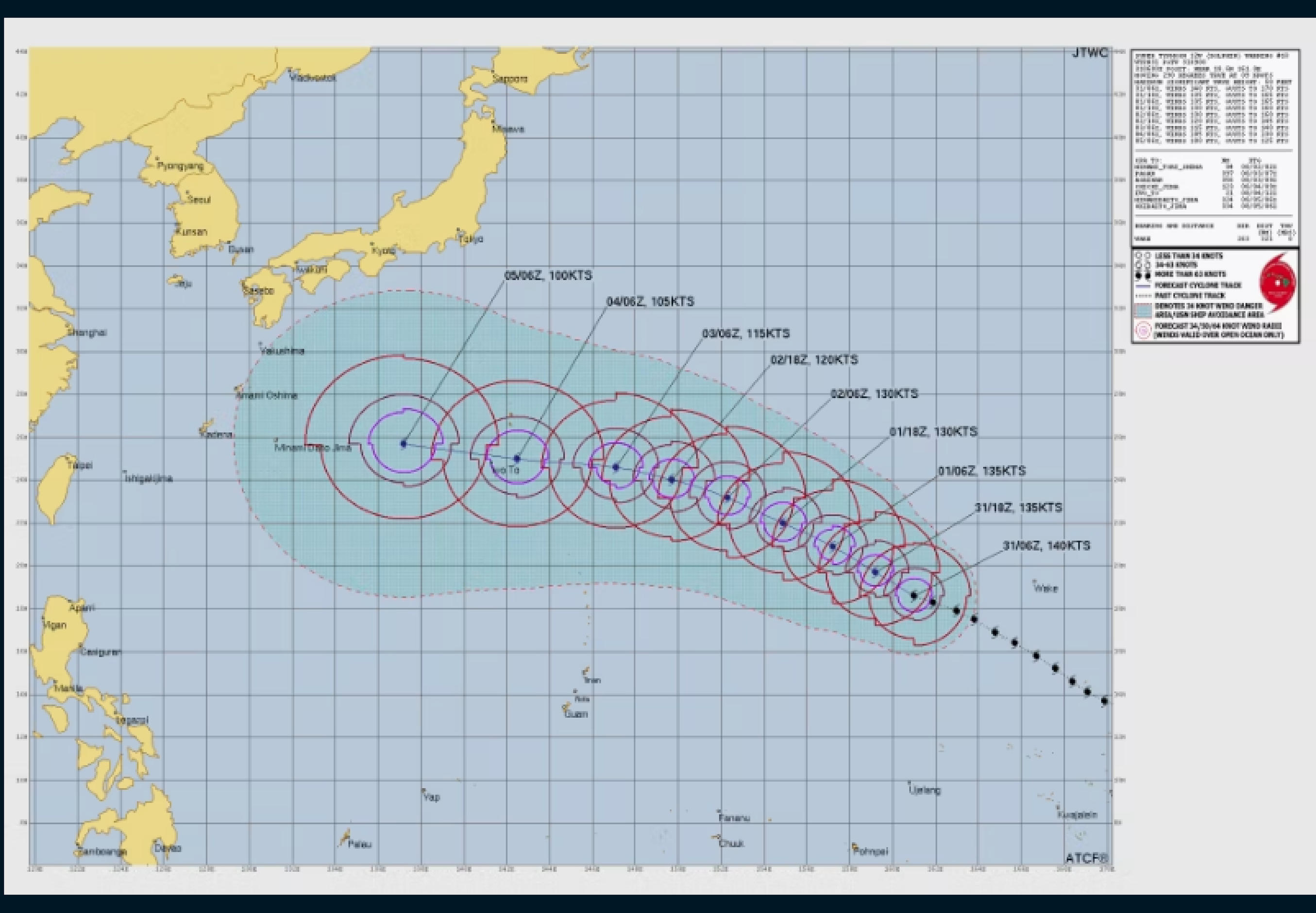


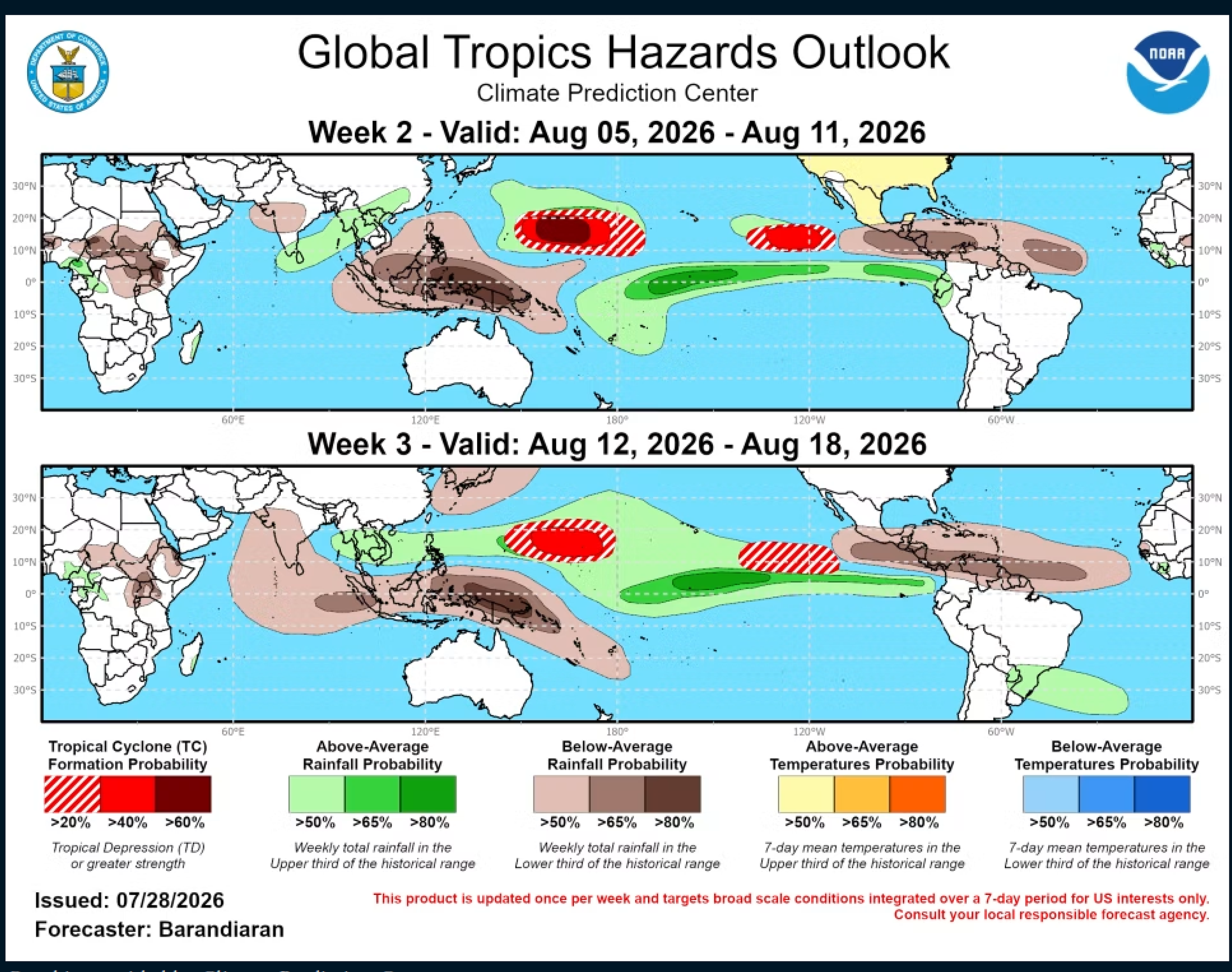


Graphic provided by Climate Prediction Center

### See my new article: *How I Met Hurricane Camille*

## What Is a Hurricane?

A hurricane (or typhoon, or severe tropical cyclone), the strongest storm on Earth, is a cyclonic (rotary) storm that derives its energy from cloud formation and rainfall, unlike frontal cyclones that derive their power from a temperature gradient.

A hurricane begins as a tropical depression with a sustained wind speed of less than 39 mph (35 knots; 63 km/hr). As the system strengthens, it becomes a tropical storm with winds from 39 to 73 mph (35-63 knots; 63-118 km/hr). Tropical storms are named in the Atlantic, East, Central and Northwest Pacific, in the South Indian Ocean, and in the Arabian Sea. When the winds are sustained (based on a one-minute average) at 74 mph (64 knots; 119 km/hr), the storm becomes: In the Atlantic Ocean, East Pacific, Central Pacific (east of the International Dateline) and Southeast Pacific (east of 160°E) a **Hurricane**; in the Northwest Pacific (west of the International Dateline) a **Typhoon**; in the Southwest Pacific (west of 160°E) and Southeast Indian Ocean (east of 90°E) a **Severe Tropical Cyclone**; in the North Indian Ocean a **Severe Cyclonic Storm**; and in the Southwest Indian Ocean (west of 90°E) a **Tropical Cyclone.**

## The Saffir-Simpson Hurricane Scale

**Category 1** – 64-82 knots (74-95 mph; 119-153 km/h). Damage is limited to foliage, signage, unanchored boats and mobile homes. There is no significant damage to buildings. The main threat to life and property may be flooding from heavy rains.

**Category 2** – 83-95 knots (96-110 mph; 154-177 km/h). Roof damage to buildings. Doors and windows damaged. Mobile homes severely damaged. Piers damaged by storm surge. Some trees blown down, more extensive limb damage.

**Category 3** – 96-112 knots (111-129 mph; 178-208 km/h). Major Hurricane. Structural damage to some buildings. Mobile homes are completely destroyed. Roof damage is common. Storm surge begins to cause significant damage in beaches and harbors, with small buildings destroyed.

**Category 4** – 113-136 knots (130-156 mph; 209-251 km/h). Structural failure of some buildings. Complete roof failures on many buildings. Extreme storm surge damage and flooding. Severe coastal erosion, with permanent changes to the coastal landscape not unheard of. Hurricane force winds extend well inland.

**Category 5** – 137+ knots (157+ mph; 252+ km/h). Complete roof failure on most buildings. Many buildings destroyed, or structurally damaged beyond repair. Catastrophic storm surge damage. In the Northwest Pacific, a typhoon that reaches 150 mph (241 km/hr) is called a Super Typhoon.

SAFFIR-SIMPSON SCALE

| Category | Knots | MPH | KM/H | Damage |
|---|---|---|---|---|
| 1 | 64-82 | 74-95 | 119-153 | Minimal |
| 2 | 83-95 | 96-110 | 154-177 | Moderate |
| 3 | 96-112 | 111-129 | 178-208 | Extensive |
| 4 | 113-136 | 130-156 | 209-251 | Extreme |
| Super Typhoon | 130+ | 150+ | 241+ | Catastrophic |
| 5 | 137+ | 157+ | 252+ | Catastrophic |

## Storm Surge

Historically, storm surge is the primary killer in hurricanes. The exact storm surge in any given area will be determined by how quickly the water depth increases offshore. In deep-water enviroments, such as the Hawaiian islands, storm surge will be enhanced by the rapidly decreasing ocean depth as the wind-driven surge approaches the coast. The peak storm surge is on the right-front quadrant (left-front in the Southern Hemisphere) of the eyewall at landfall, where on-shore winds are the strongest, and at the leading edge of the eyewall. Contrary to a popular myth, the storm surge is entirely wind-driven water—it is not caused by the low pressure of the eye. Another factor in the severity of the storm surge is tide. Obviously, an 18-foot storm surge at high tide is that much worse than an 18-foot surge at low tide.

## Tropical Cyclone Formation

*By Jonathan Edwards*

Tropical Cyclone Genesis is the technical term for the process of storm formation that leads ultimately to what are called hurricanes, typhoons, or tropical cyclones in various parts of the world.

This occurs when, in the Northern Hemisphere, the Intertropical Convergence Zone, or ITCZ, shifts northward out of the doldrums and atmospheric conditions become favorable for tropical cyclone formation after about the middle of May.

A series of low-pressure ripples develops within the ITCZ. These are known as tropical waves and progress from east to west. In the late season, they typically shift their movement toward the west-nothwest, or even northwest, after crossing 45° or 50° W longitude.

These tropical waves, ideally embedded in the deep-layer easterly flow, contain a northeast wind shift. This is typically referred to as a "convergence", where lines of equal atmospheric pressure are pressed together between the high-pressure ridge to the north and the developing low-pressure system. The divergence that results ahead of the convergence zone gives us a notheasterly wind as the axis of the tropical wave approaches. Gusts up to 25 mph may occur. Sometimes there can be gusts to tropical storm force in stronger waves. There can be next to no weather associated with these waves, and they may pass virtually unnoticed. More typically, there are bands of disturbed weather riding the axis of the wave. *Read more …*

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